

Trained



train



Large labelled
training set

Environment-specific
features

A

Standard



Cross-env.



Training-free, prior work



Frozen patch
features



Codebook



Segmentation

CLS attention

Prior-selected features

B

Standard



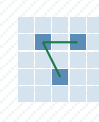
Cross-env.



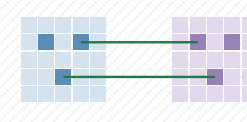
Ours



Frozen patch
features



Within-image
Balanced, matchable patch evidence



Across-image
Balanced, matchable patch evidence

C

Standard

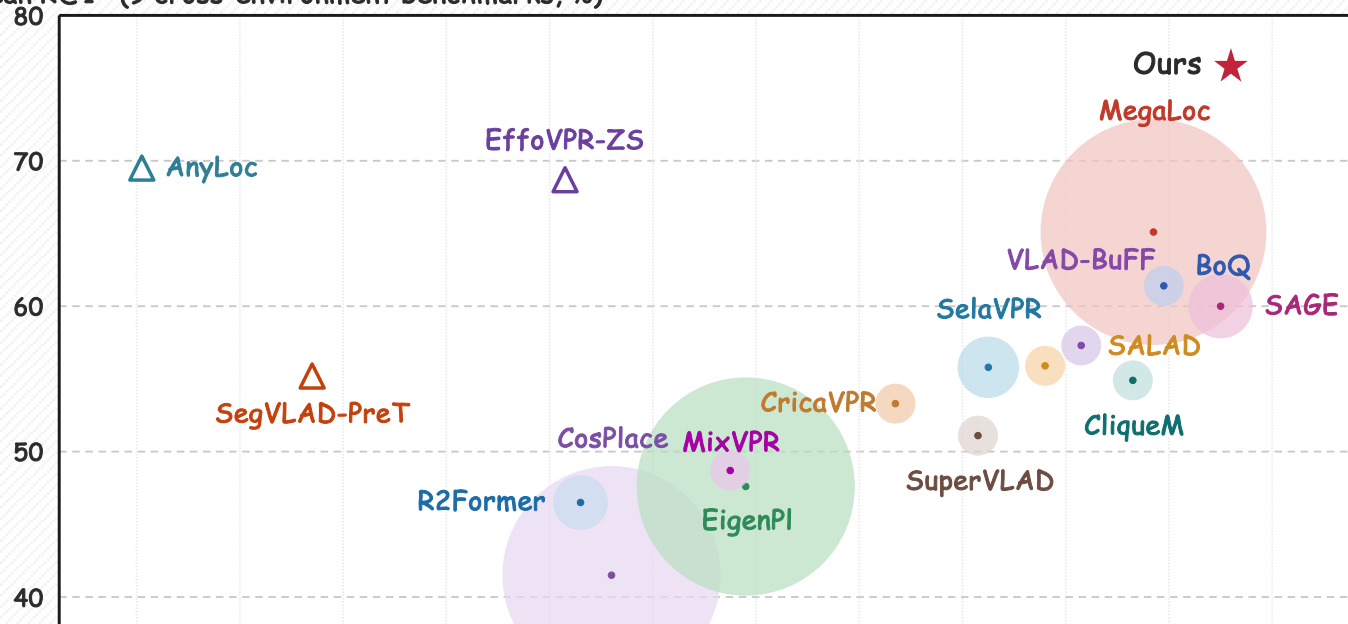


Cross-env.

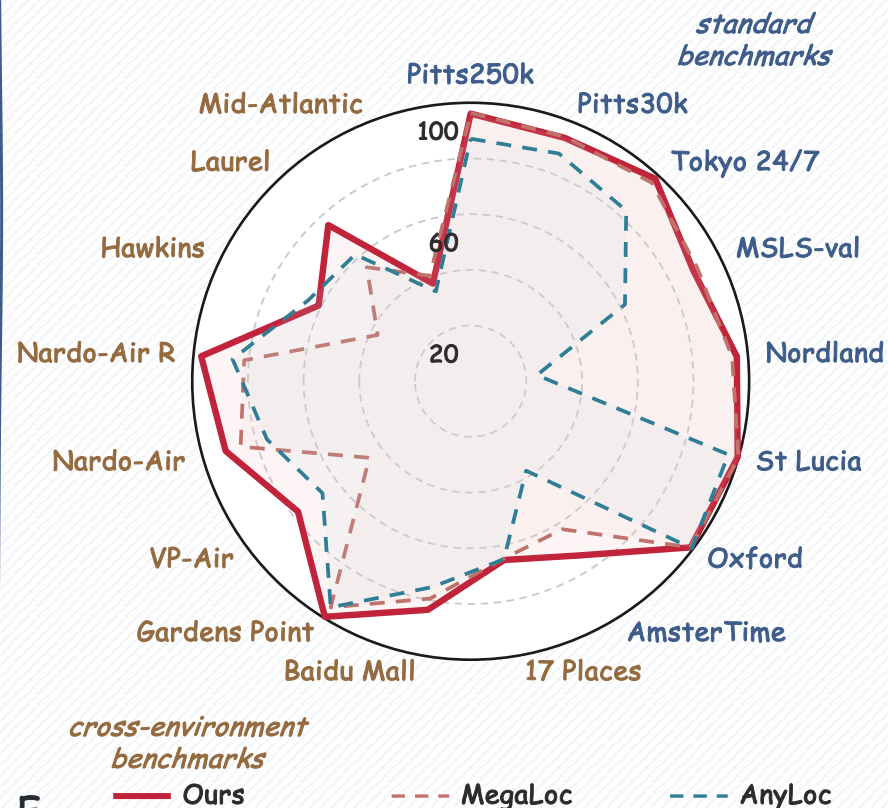


Training cost vs. performance

mean R@1 (9 cross-environment benchmarks, %)



Performance on 17 benchmarks



cross-environment
benchmarks

Ours

MegaLoc

AnyLoc

E

★ ours (training-free) △ training-free ● trained ● 1 million images ● 5 million images

D